experiment progress manager agent that coordinates four clearly defined stages of scientific experimentation:

Stage 1 **Preliminary Investigation**: Establishing initial feasibility and correctness through a minimal working prototype based on the generated research idea.
Stage 2 **Hyperparameter Tuning**: Refining the initial implementation by optimizing critical hyperparameters (e.g., learning rate, epochs) to create a robust experimental baseline.
Stage 3 **Research Agenda Execution**: Systematically implementing the core research agenda based on the tuned baseline.
Stage 4 **Ablation Studies**: Systematically assessing the importance of various research components, providing rigorous support for the main experimental findings.

Each stage has explicit stopping criteria. Stage 1 concludes when a basic working prototype is successfully executed. Stage 2 ends when experiments stabilize, as indicated by convergence in training curves and successful execution across at least two datasets. Stages 3 and 4 conclude when the allocated computational budget is exhausted. Stage 3 also includes a check for experiment duration–if runs finish much faster than the pre-allocated runtime, the system suggests increasing the complexity of experiments.

After each stage, the experiment manager selects the best-performing node using a dedicated LLM evaluator (see next section) based on clearly articulated criteria. This selected node is then carried forward to seed the subsequent experimentation stage. The manager also records checkpoints at each stage's completion. To ensure scientific rigor and reproducibility, the experiment manager launches multiple replications of the selected best experiments at the conclusion of each stage. These repeated runs provide statistics (mean and standard deviation) for figures and reported results.

3.2.2. Parallelized Agentic Tree Search

THE AI SCIENTIST-V1 operated strictly linearly, where each code refinement directly built on the immediately preceding experiment. In contrast, THE AI SCIENTIST-V2 adopts a significantly more flexible and exploratory approach inspired by recent successes in integrating tree search with LLM-driven workflows (Chan et al., 2025; Jiang et al., 2025; Wijk et al., 2024) and research on open-endedness (Clune, 2019; Mouret and Clune, 2015). We incorporate this agentic tree search approach across all four experimentation stages outlined in §3.2.1, enabling deeper and more systematic exploration of scientific hypotheses.

Each experimental node within our tree-based framework undergoes the following execution cycle: An LLM first generates both a concrete experimentation plan and the associated Python code to implement the experiment. The generated code is immediately executed in a Python interpreter. If execution encounters an error, the error message is recorded, and the node is marked as **buggy**, ending the current execution cycle for that node. If execution succeeds, the experiment proceeds to the *plotting phase*.

During each experiment, the system is instructed to save all relevant experimental outputs (training and validation metrics, losses, etc.) into structured numpy files. In the plotting phase, THE AI SCIENTIST-V2 reads these stored results and the code, generating visualizations that summarize and illustrate the findings clearly. These visualizations are subsequently passed to a Vision-Language Model (VLM) for critique. Any issues flagged by the VLM (such as unclear labels, missing legends, or misleading visualizations) result in the node being marked as **buggy**, and this feedback is recorded for future debugging. Nodes that successfully execute and pass the VLM review without issue are designated as **non-buggy**.

We define each node as a collection comprising an experiment script (e.g., a Python file), a textual